

Pathophysiology of Disease Progression in Proteinuric and Nonproteinuric Kidney Disease

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Chronic kidney disease (CKD) is a worldwide threat to public health and has a risk-multiplier effect on major noncommunicable diseases, including cardiovascular diseases.¹ More than 697 million individuals are currently affected by CKD worldwide,² and the number of patients with end-stage kidney disease (ESKD) treated with renal replacement therapy with dialysis or transplantation globally exceeds 2.6 million people.³ Independent of the initial insult, progression to ESKD is relatively common in chronic nephropathies. Many forms of progressive noncystic kidney disease are glomerular in origin, and yet it is the intensity of the accompanying or evolving injury of the tubulointerstitial compartment, rather than the extent of glomerular changes, that predicts the overall decline in kidney function.⁴ Although genetic factors contribute to susceptibility and progression of kidney disease, increased glomerular capillary flow and pressure consistently leading to increased urinary protein traffic have been claimed as independent factors of progression and poor kidney outcomes in nondiabetic and diabetic kidney disease.^{5,6} The Ramipril Efficacy in Nephrology (REIN) study was the first trial to formally test the role of proteinuria in the progression of kidney disease.⁷⁻⁹ The trial showed that in 352 patients with proteinuric nephropathies from different etiologies, higher proteinuria at inclusion was associated with faster glomerular filtration rate (GFR) decline and progression of ESKD on follow-up.^{5,10} Notably, in the REIN trial, larger proteinuria reduction and less residual proteinuria at follow-up were both associated with slower GFR decline and more effective protection against progression to ESKD, independent of treatment allocation.¹¹ Greater proteinuria reduction predicted slower progression, as in the patients with type 2 diabetes with overt nephropathy included in the Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan (RENAAL)⁶ study and Irbesartan Diabetic Nephropathy Trial (IDNT).¹² The predictive pathogenic role of proteinuria was confirmed by a pooled analysis of 2387 CKD patients included in 11 trials, which showed that, irrespective of treatment, short-term changes in proteinuria were strongly consistent with long-term outcomes, whereas no effect on proteinuria predicted no long-term benefit.¹³ Thus, efforts to dissect the mechanisms and mediators underlying disease progression in proteinuric nephropathies are of utmost importance to help design novel medications to further improve the efficacy of current renoprotective interventions.

FROM GLOMERULAR HEMODYNAMIC CHANGES TO LOSS OF CHARGE- AND SIZE-SELECTIVE PROPERTIES

Under normal conditions, the kidney filtration barrier prevents the passage of macromolecules from the blood into the urinary space. The ability to retain proteins in the glomerular capillary is influenced by the intrinsic permeability and selectivity properties of the three distinct but interacting layers: fenestrated glomerular endothelial cells,

the glomerular basement membrane (GBM), and podocytes with their foot processes and slit diaphragms.¹⁴⁻¹⁶ The glomerular barrier properties allow for high filtration rate of water and nonrestricted passage of small and middle-size molecules while maintaining charge and size selectivity for serum albumin and larger proteins.

The luminal surface of glomerular endothelial cells is covered by the glycocalyx, a thin, negatively charged layer of membrane-bound glycoproteins, proteoglycans, and glycosaminoglycans.¹⁷ The negative charges prevent the transmural passage of anionic circulating proteins, such as albumin, by acting as an electrostatic barrier. Therefore, although endothelial fenestrae are much larger than albumin (about 60 nm in diameter compared with the albumin's radius of only 3.6 nm), negatively charged circulating macromolecules stay away from the endothelial surface due to electrical repulsion and remain within the circulation. It is now evident that the first restriction to albumin filtration across the glomerular capillary wall consists in the endothelial surface layer, and its role is to substantially decrease protein concentration in the fluid that enters the GBM layer.¹⁷ The thickness of the glycocalyx layer appears to be importantly affected by shear stress. In particular, increased fluid shear stress is associated with glycocalyx formation and reorganization on the cell surface in contact with fluid flow.¹⁸ Thus, pathologic conditions in which changes in glomerular capillary flow (i.e., reduced flow) may occur are expected to decrease glycocalyx formation, thereby reducing the retention of anionic proteins within the bloodstream. It has been demonstrated that disruption of the endothelial glycocalyx increases glomerular albumin filtration even in the presence of only minor changes in both the GBM and glomerular epithelial cells.

The epithelial filtration slit of podocytes is a major component of the glomerular barrier that also contributes to restrict protein transport. As shown from animal models in the early 1980s, a variety of insults ultimately lead to a common pathway of glomerular capillary hyperperfusion and hypertension, followed by increased permeability with excess passage of proteins across the glomerular capillary wall and progressive glomerular injury.¹⁹ Enhanced intraglomerular capillary pressure stretches the glomerular wall, which, in addition to directly injuring glomerular cells,²⁰ impairs the selective function of the glomerular capillary, an effect explained by the appearance of very large pores that exceed the sizes observed in normal conditions and allow increased filtration of plasma proteins (Fig. 81.1).²¹ Mechanical strain also increases angiotensin II (Ang II) production and expression of Ang II type 1 (AT₁) receptors in podocytes.²² Independent of its hemodynamic effects, Ang II may directly impair the glomerular barrier sieving function, possibly through inhibition of podocyte expression of nephrin, the essential protein component of the glomerular slit diaphragm (Fig. 81.2).²³ This observation has been confirmed in studies in diabetic animals showing that blockade of Ang II synthesis/activity preserved the expression of nephrin in the glomeruli and prevented

Proteinuric Rats Have Large Pores in the Slit Diaphragm, Allowing the Passage of Proteins

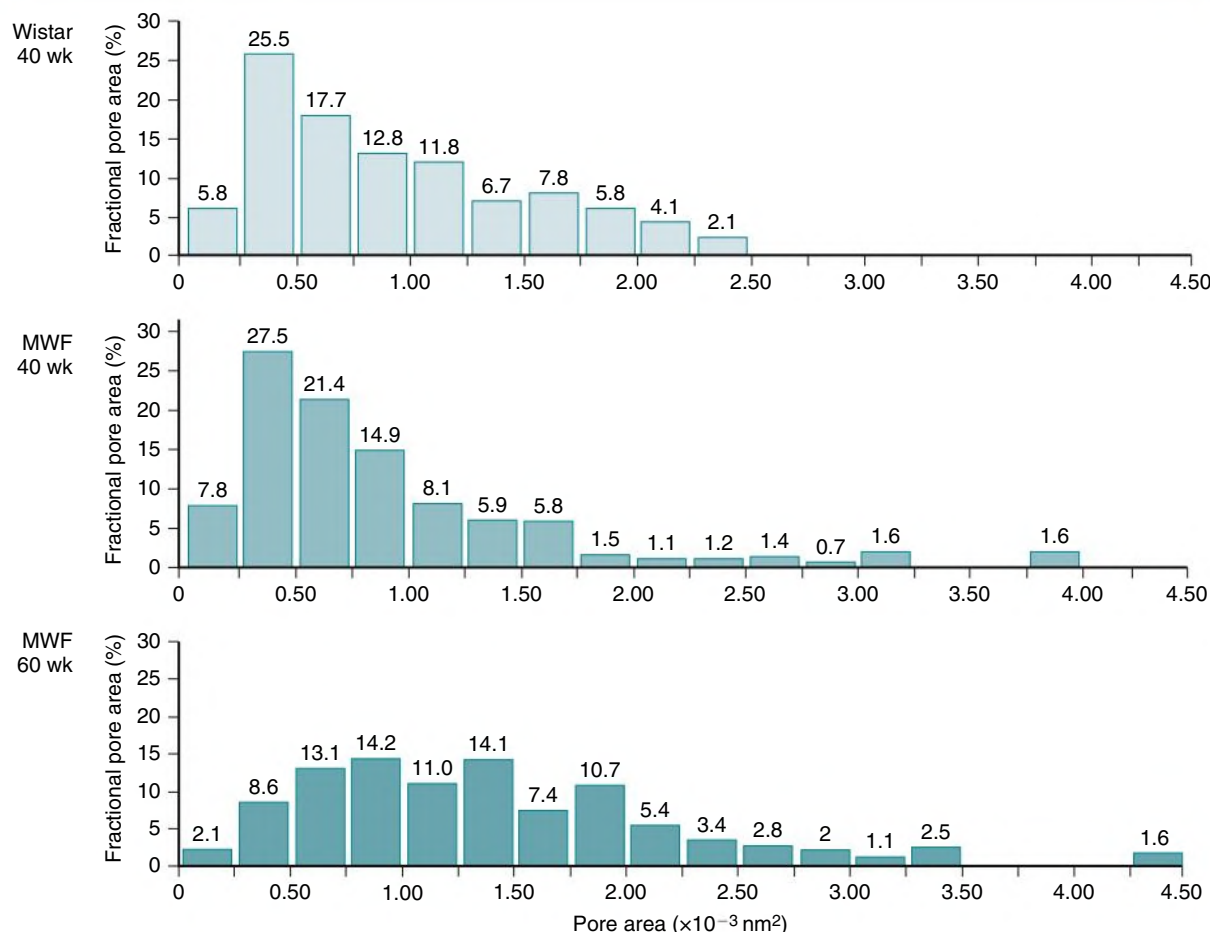


Fig. 81.1 Proteinuric Munich Wistar Frömter (MWF) Rats Have Large Pores in the Glomerular Capillary Slit Diaphragm That Exceed the Size Observed in Normal Animals. Distribution of fractional pore area (the fraction of the total pore area for given pore area interval) calculated for slit pores in 40-week-old normal Wistar rats and in 40- or 60-week-old proteinuric MWF rats. (From Gagliardini E, Conti S, Benigni A, Remuzzi G, Remuzzi A. Imaging of the porous ultrastructure of the glomerular epithelial filtration slit. *J Am Soc Nephrol*. 2010;21[12]:2081–2089.)

overt proteinuria.²⁴ Thus, in the setting of diabetes, after the initial insult of hyperglycemia and intraglomerular hypertension, Ang II plays a relevant role in sustaining glomerular injury via persistent activation of Notch1 and Snail signaling in the podocyte, eventually resulting in persistent downregulation of nephrin expression.²⁵ The consistency of the findings in Zucker diabetic fatty rats with overt nephropathy and in patients with type 2 diabetes and established nephropathy provides a robust reason to infer an important role for the Ang II Notch1/Snail axis in perpetuating loss of glomerular size-selective properties. A recent experimental study points to protective actions of mitochondrial glycerol 3-phosphate dehydrogenase (mGPDH), a protein located on the inner mitochondrial membrane, against podocyte injury and progression of diabetic kidney disease.²⁶ In particular, mGPDH expression was found to be downregulated in podocytes of mice and patients with diabetic nephropathy. Podocyte-specific mGPDH knockout mice showed exacerbated diabetes-induced proteinuria, podocyte damage and glomerulosclerosis. Inhibition of the receptor for the advanced glycation end-product (RAGE) abrogated mGPDH-induced podocyte injury and glomerular disease, suggesting that the effects of this mitochondrial protein were exerted through the regulation of the

RAGE signaling pathway. Restoration of mGPDH expression in mice with established glomerular injury resulted in renoprotection, making mGPDH a potential therapeutic target for ameliorating diabetic kidney disease.²⁶

Given the abundant evidence that podocyte injury underlies most, if not all, proteinuric kidney diseases, new technologies, including ultrahigh-resolution imaging and genetically engineered models of human disease, were used to develop a biophysical theoretical model of glomerular ultrafiltration.^{27,28} According to this model, remarkable intraluminal hydrostatic pressure exerts physical forces on the capillary wall that are counteracted by the GBM and by podocytes. When podocytes are injured, they take on a simplified, effaced phenotype, and the slit diaphragm length is much reduced. Concomitantly, the capacity of podocyte foot processes to provide compressive forces to counteract a transmembrane pressure of 40 mm Hg is decreased, thereby reducing the compression of GBM, with eventual increase in the permeability to albumin.^{27,28} However, this hypothesis does not account for the role played by the subpodocyte space (SPS) which surrounds the filtration membrane and increases under pathologic conditions.^{29,30} Indeed, evidence from studies that used Munich Wistar Frömter (MWF) rats, an

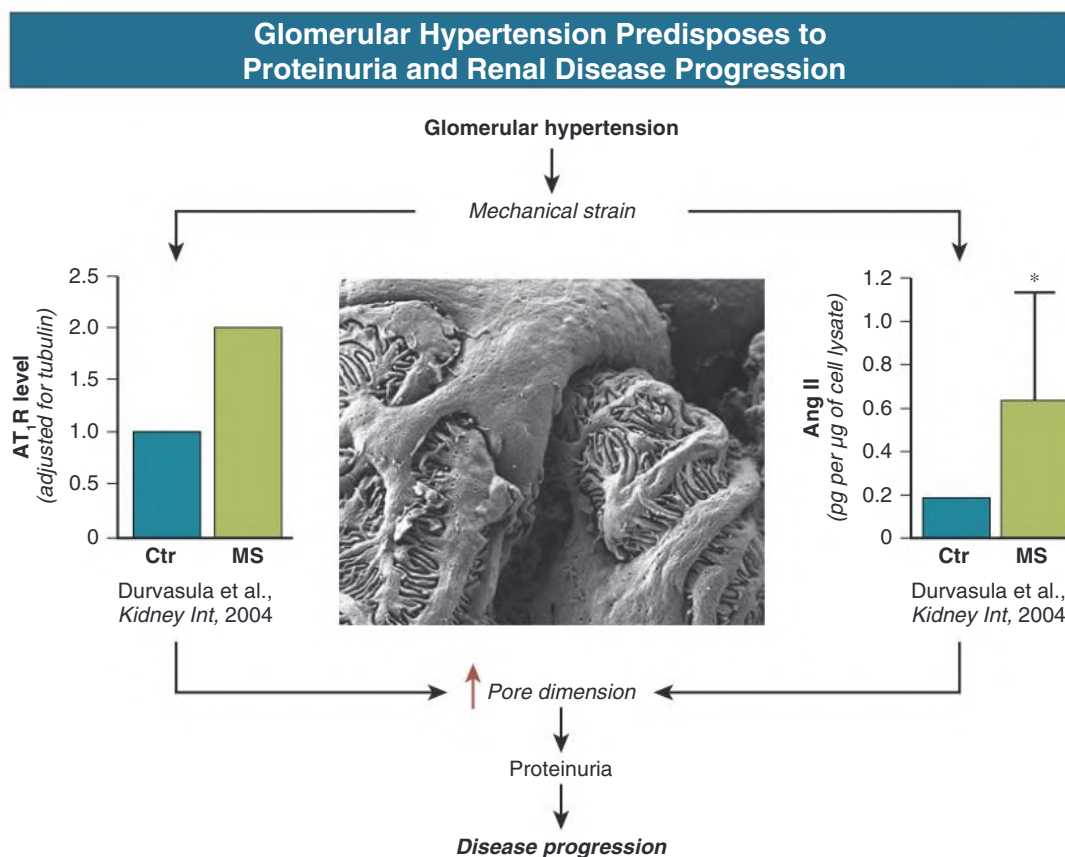


Fig. 81.2 Schematic Process Linking Glomerular Capillary Hypertension to Proteinuria and Kidney Disease Progression. After the original disease process causes an initial loss of nephron units, enhanced intraglomerular capillary pressure stretches the glomerular wall, impairing its size-selective function. Mechanical strain also increases angiotensin II (Ang II) production and expression of Ang II type 1 receptors (AT₁R) in podocytes, eventually directly contributing to loss of the glomerular barrier sieving function through inhibition of podocyte slit diaphragm nephrin expression. **P* < .05 vs control. Ctrl, Control; MS, mechanical strain. (From Durvasula RV, Petermann AT, Hiromura K, et al. Activation of a local tissue angiotensin system in podocytes by mechanical strain. *Kidney Int*. 2004;65[1]:30–39.)

experimental model of spontaneous proteinuria and glomerulosclerosis,³¹ indicates that loss of glomerular filtration function depends strongly on abnormal rearrangement of podocytes, which causes an expansion of the SPS, an increase in its hydraulic resistance to filtered water, and an increased force acting within the SPS, which tends to detach podocytes from the capillary membrane.³² This evidence is further corroborated by the finding that angiotensin-converting enzyme (ACE) inhibition preserves podocyte structure, reducing the spatial extension of SPS and restoring the hydrodynamic detaching forces which act on podocytes, eventually normalizing water filtration and glomerular barrier function.³²

Podocyte Response to Protein Trafficking

Besides being affected by mechanical stress, podocytes are also damaged by excessive protein load resulting from alteration of glomerular permeability to macromolecules.³³

Protein uptake by podocytes may occur through binding to megalin, a receptor for albumin and immunoglobulin light chains that is endocytosed after ligand binding, as shown in cultured murine podocytes.³⁴ Excessive protein uptake by podocytes also induces transforming growth factor- β (TGF- β) production, which contributes to cell apoptosis, an additional cause of podocyte loss in proteinuric glomerulopathies.³⁵

Finally, podocytes possess a fully functional endothelin (ET) system, and evidence has highlighted the role of ET-1 in promoting

structural and functional alteration of these cells in kidney disease.³⁶ This possibility is supported by the beneficial effects of ET receptor antagonists on the development of proteinuria and glomerular injury in type 1 diabetic animals, partially attributed to attenuation of podocyte loss.²⁴ Although the mechanism of podocyte preservation by ET receptor antagonism remains ill defined, recent observations indicate that ET-1 induces an epithelial-to-mesenchymal transition-like event associated with increased podocyte motility.³⁷ This is dependent on the activation of ET_A receptor, which recruits β -arrestin 1, leading to epithelial growth factor receptor translocation and β -catenin phosphorylation, in turn promoting the expression of migratory genes.³⁷

Furthermore, the preservation of podocyte structure under ET receptor antagonism could be secondary to a reduction in the protein load reaching the cells, as suggested by recent findings in diabetic apoE knockout mice.³⁸ In this model, the ET_A receptor antagonist atrasentan restored the structural integrity of glomerular endothelial glycocalyx normalizing the barrier against the excessive traffic of albumin through the capillary wall up to podocytes.³⁸

Crosstalk of Podocytes With Mesangial and Endothelial Cells

The mesangial cells are a critical part of the glomerular functional unit, interacting closely with podocytes.³⁹ Alterations in one cell type can produce changes in the others. Whether cytokines generated by podocytes influence mesangial cells has yet to be clearly defined, but

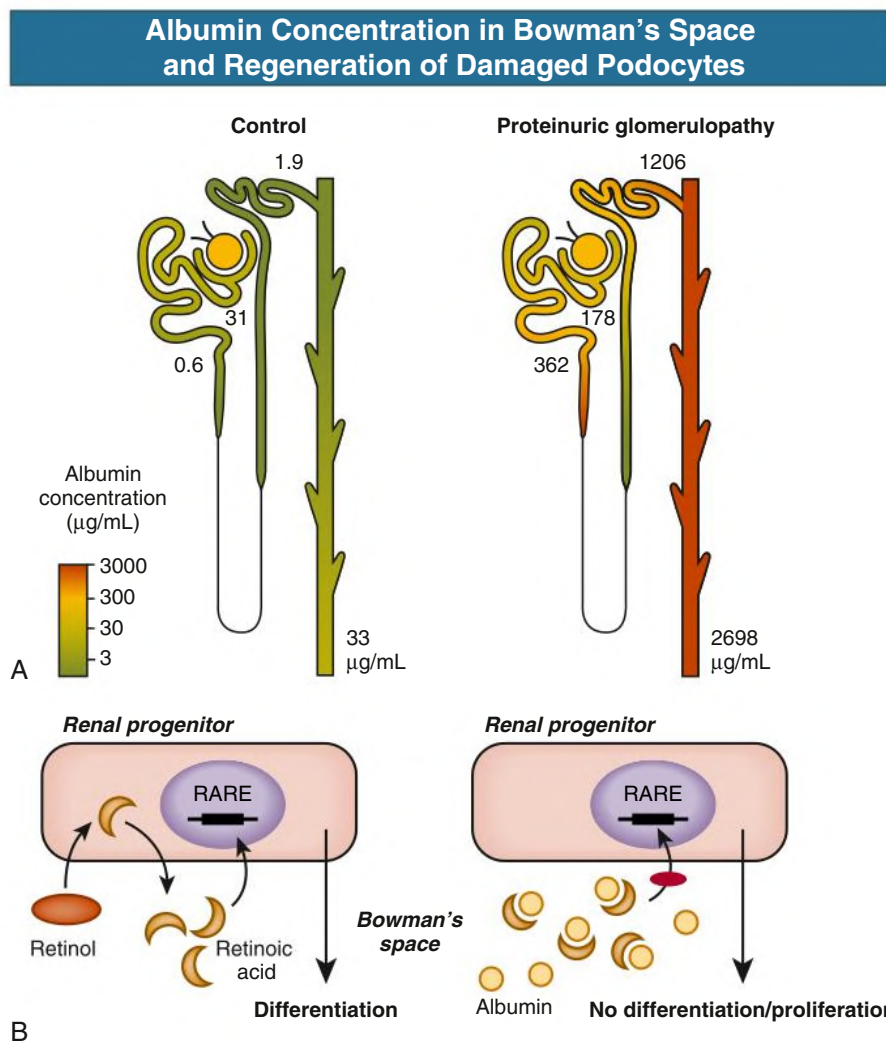


Fig. 81.4 Albumin Concentration in Bowman's Space and Regeneration of Damaged Podocytes. (A) In the proteinuric glomerulopathies, changes in glomerular perm-selective function results in a high concentration of albumin in Bowman's space. Color-coded representation of estimated albumin concentration along the entire nephrons in two animal groups (control and kidney mass ablation model). Numbers represent local group average albumin concentration in micrograms per milliliter. (B) In vitro albumin overload, prevented the differentiation to podocytes from human kidney progenitor cells by sequestering retinoic acid, thus impairing retinoic acid response element (RARE)-mediated transcription of podocyte-specific genes. After early podocyte injury, retinol is lost through the injured glomerular filtration barrier and transformed into retinoic acid. Albumin overload also promotes proliferation and migration of kidney progenitor cells into Bowman's space. (A, Modified from Sangalli F, Carrara F, Gaspari F, et al. Effect of ACE inhibition on glomerular permselectivity and tubular albumin concentration in the renal ablation model. *Am J Physiol Renal Physiol*. 2011;300[6]:F1291–F1300. B, Modified from Peired A, Angelotti ML, Ronconi E, et al. Proteinuria impairs podocyte regeneration by sequestering retinoic acid. *J Am Soc Nephrol*. 2013;24[11]:1756–1768.)

claudin⁺ PECs increased in number, proliferated, and accumulated in Bowman's space, whereas parietal podocytes decreased. These findings suggested dysregulation of PECs' ability to differentiate into podocytes and to repair injury.⁵⁰

PECs expressing the progenitor cell marker also have been reported to proliferate and accumulate into the multilayered cellular lesions in patients with glomerulonephritides characterized by extracellular capillary proliferation.⁵² Upregulation of the CXCR4 chemokine receptor on these progenitor cells was found to be accompanied by high expression of its ligand, SDF-1, in podocytes.⁵² Moreover, PEC proliferation was associated with increased expression of the AT₁ receptor. Renin-angiotensin system blockade normalized CXCR4 and AT₁ receptor expression on parietal progenitor cells concomitant with regression of

crescentic lesions.⁵² Together these findings suggest that the glomerular hyperplastic lesions derive from the proliferation and migration of kidney progenitor cells in response to injured podocytes and that the Ang II/AT₁ receptor pathway may contribute, together with SDF-1/CXCR4 axis, to the dysregulated response of PEC precursors. A recent study highlighted the key role of complement components C3 and C3a in determining podocyte dysfunction and loss leading to PEC activation.⁵³ Proliferation and migration of PECs in response to activated complement contribute to the development of glomerular sclerotic lesions.⁵³ Consistently, studies on kidney biopsy samples from patients with proteinuric nephropathies and PEC activation showed concomitant glomerular C3 and C3a deposition, indicating its key role in the development of glomerular lesions.⁵³

PROXIMAL TUBULAR CELL INJURY

Glomerular ultrafiltration of excessive amounts of plasma protein—associated factors incites tubulointerstitial damage and further promotes the effects of glomerular disease on the tubulointerstitial compartment.

The harmful substances in the proteinuric ultrafiltrate may incite tubular epithelial injury with tubular apoptosis, secondary generation of inflammatory mediators, and peritubular inflammation.⁴¹ The mechanisms by which increased urinary protein concentration leads to toxic injury are multifactorial and involve complex interactions among numerous pathways of cellular damage.

Tubular Cell Apoptosis

Kidney proximal tubular cells have a remarkable ability to reabsorb large quantities of albumin through clathrin- and megalin receptor-mediated endocytosis.⁵⁴ Kidneys in rats with albumin overload proteinuria⁵⁵ or passive Heymann nephritis (PHN)⁵⁶ showed increased numbers of terminal deoxynucleotidyl transferase nick-end labeling-positive apoptotic cells in the tubulointerstitial compartment. In tubules, most of the positive cells expressed Ang II type 2 (AT₂) receptors.⁵⁵ Findings of reduced phosphorylation of extracellular signal-regulated kinase and Bcl-2 suggested an AT₂ receptor-mediated mechanism underlying tubular cell apoptosis.⁵⁵ Similarly, apoptotic cells expressing both proximal and distal tubular phenotypes were

detected in biopsy specimens from patients with primary focal segmental glomerulosclerosis (FSGS), and a positive correlation was documented between proteinuria and incidence of tubular cell apoptosis.⁵⁷

Proximal tubular cell apoptosis has been shown to contribute to tubuloglomerular disconnection and atrophy in response to proteinuria in animal models of proteinuric nephropathies.⁵⁶ When passive PHN rats were given the ACE inhibitor lisinopril, which limits proteinuria, tubular atrophy and disconnection were remarkably prevented.⁵⁶ PHN animals that did not respond to ACE inhibition in terms of reduction of proteinuria had a glomerular population consisting mainly of atubular glomeruli and glomeruli connected with atrophic tubules, which again is consistent with the possibility that excessive protein tubular handling favors disconnection.⁵⁶

Tubular Cell Phenotypic Changes

Excessive protein uptake at the apical pole of the proximal tubular cells is associated with phenotypic changes characteristic of an activated state.

Insights into specific mechanisms linking protein uptake to cell activation have come from in vitro studies using polarized tubular cells to assess the effect of apical exposure to proteins. Collectively, they show that protein overload induces a proinflammatory phenotype (Fig. 81.5).⁵⁸ Indeed, upregulation of inflammatory and fibrogenic genes and production of related proteins have been reported after a challenge of proximal tubular cells with plasma proteins. They

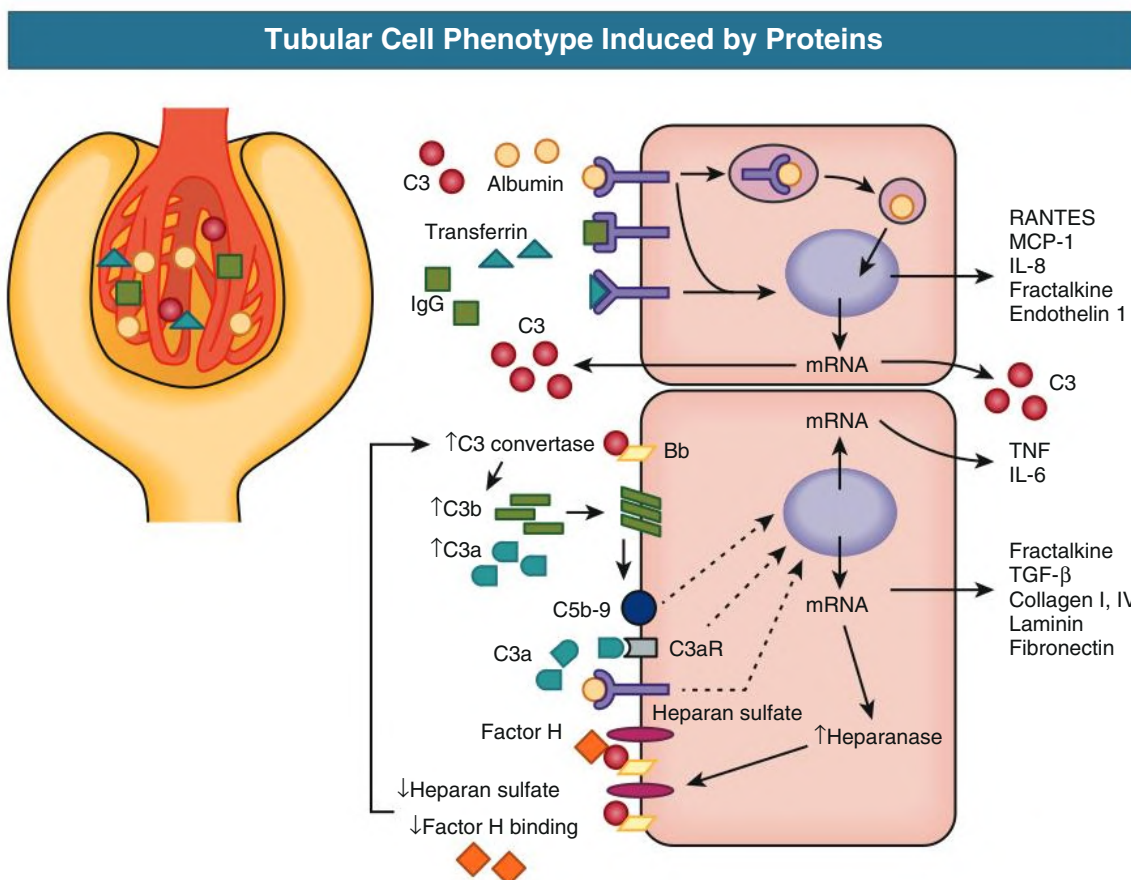


Fig. 81.5 Tubular Cell Phenotypic Changes Induced by Proteins. Protein overload of proximal tubular cells as a consequence of increased glomerular permeability to proteins activates intracellular signals that promote cell apoptosis or cause increased production of inflammatory and vasoactive mediators and growth factors. *IgG*, Immunoglobulin G; *IL-6*, interleukin-6; *IL-8*, interleukin-8; *MCP-1*, monocyte chemoattractant protein-1; *RANTES*, regulated upon activation, normal T cell expressed and secreted; *TGF-β*, transforming growth factor-β; *TNF*, tumor necrosis factor.

include cytokines, chemokines, and vasoactive substances, such as monocyte chemoattractant protein-1 (MCP-1); regulated upon activation, normal T cell expressed and secreted (RANTES); interleukin-8; fractalkine; and ET-1.⁵⁸ Moreover, levels of the profibrogenic cytokine TGF- β and its type I receptor, tissue inhibitors of metalloproteinase (TIMP-1 and TIMP-2), as well as membrane surface expression of the $\alpha\beta 5$ integrin, were also highly increased in vitro on stimulation by plasma proteins.⁵⁸ Investigation of the molecular mechanisms underlying chemokine and growth factor upregulation in proximal tubular cells on protein challenge has focused on the activation of the transcription factor nuclear factor- κ B (NF- κ B).⁵⁸ In this context, a recent study showed that albumin is capable of inducing chromatin modification, making the promoter region of the gene encoding miR-184, which plays a key role in tubulointerstitial nephritis of diabetic rats, more accessible to NF- κ B inducing the miR-184 transcription.⁵⁹ Other studies confirmed the role of the NF- κ B pathway and revealed reactive oxygen as a second messenger.⁶⁰ The link of excessive protein reabsorption to tubular cell activation has been confirmed by in vivo studies in animal models. Evidence of early activation of proximal tubule cells during nonselective proteinuria was derived from experiments in megalin knockout/NEP25 mice treated with the immunotoxin LMB2, a model of nephrotic syndrome, FSGS, and tubulointerstitial injury.⁶¹ Megalin-deficient proximal tubule cells reabsorbed fewer proteins in vivo and expressed fewer tubular injury markers, such as MCP-1 and heme-oxygenase.⁶¹ Moreover, a report using podocin knockout mice as a model of FSGS demonstrated that increased reabsorption of abnormally filtered albumin and other proteins in proximal tubule cells was associated with massive increase of lysosomal proteolysis.⁶² However, high expression of proinflammatory and profibrotic mediators (MCP-1 and TGF- β) in the kidney also occurred almost simultaneously with the increase in tubular content of megalin ligands.⁶² This suggests that, despite the effective degradation of endocytosed proteins by lysosomes in this model, the excessive protein uptake by tubule cells was still capable of activating phenotypic changes by upregulation of proinflammatory and growth factor encoding genes.

With proteinuria, a putative key factor in tubular cell activation and damage is the excess glomerular filtration of plasma-derived complement C3, the central molecule in the complement system that exerts proinflammatory potential. Kidney tubule epithelial cells appear more susceptible to luminal attack by the C5b-9 membrane attack complex because of the relative lack of membrane-bound complement regulatory proteins, such as membrane cofactor protein (CD46), decay-accelerating factor, or CD55 and CD59 on the apical surface.⁶³ In rats with reduced kidney mass, C3 colocalized with proximal tubular cells engaged in high protein uptake. By limiting the transglomerular passage of proteins, treatment with ACE inhibitors was an effective manner to reduce the C3 load of tubular cells in remnant kidneys.⁶⁴ C3 and other complement proteins are also found in proximal tubules in kidney biopsy material from patients with nephrosis.⁶⁵ Furthermore, proximal tubular cells are able to synthesize C3 and other complement factors and to upregulate C3 in response to serum proteins in vitro.⁶⁶ Studies in C3-deficient kidneys transplanted into wild-type mice have helped clarify that ultrafiltered C3 contributes more to tubular damage induced by protein overload than locally synthesized C3.⁶⁷

INTERSTITIAL INFLAMMATION AND FIBROSIS

In proteinuric kidney disease, progressive inflammation and injury to the kidney interstitium are secondary events after glomerular or vascular damage. Cytokines and chemokines synthesized by proximal

tubular epithelial cells are released into the interstitium, where they contribute to recruit inflammatory cells and lymphocytes causing progressive fibrosis.⁵⁸

Resident Monocyte and Lymphocyte Activation

The interstitium of normal kidneys contains numerous resident monocytic myelocytes, which express dendritic cell markers and can present antigens.⁶⁸ Dendritic cells have been described to form an immune sentinel network through the entire kidney, where they probe the environment in search of antigens.⁶⁹ An inflammatory environment converts the tolerogenic status of resident dendritic cells into an immunogenic one, favoring recruitment of T cells. The importance of kidney dendritic cell activation to kidney injury has been demonstrated by the fact that in transgenic NOH mice (which selectively express the model antigens ovalbumin and hen egg lysozyme in podocytes), dendritic cell depletion resolved established periglomerular mononuclear infiltrates.⁷⁰ In vitro experiments also have shown that exposure of rat proximal tubular cells to excess autologous albumin, as in the case of proteinuric nephropathies, results in the formation of the N-terminal 24-residue fragment of albumin (ALB₁₋₂₄).⁷¹ This peptide is taken up by dendritic cells, where it is further processed by proteasomes into antigen peptides. These peptides were shown to have the binding motif for major histocompatibility complex (MHC) class I and to be capable of activating CD8⁺ T cells. Moreover, in vivo, in the rat model of kidney mass ablation, accumulation of dendritic cells in the kidney parenchyma peaked 1 week after surgery and decreased thereafter, concomitant with their appearance in the draining lymph nodes. Dendritic cells from kidney lymph nodes loaded with the albumin peptide ALB₁₋₂₄ activated syngeneic CD8⁺ T cells in primary culture.⁷¹ Thus, inflammatory stimuli released from damaged tubules after protein overload may represent danger signals that, in the presence of albumin peptides, alert dendritic cells to promote local immune response via CD8⁺ T cells, which are activated in regional lymph nodes and recruited in the kidney interstitium (Fig. 81.6).

The interstitium infiltrate in most human chronic kidney diseases consists of a number of different effector cells, including macrophages and CD4⁺ T cells, in addition to CD8⁺ T cells.⁷² However, there is substantial functional diversity among CD4⁺ T cells, so that certain subpopulations, such as CD4⁺CD25⁺ T cells (T regulatory [Treg] cells), hinder rather than help the immune response.⁷³ Findings of a study using the green fluorescence protein *Foxp3* mouse suggest that *Foxp3* expression identifies the regulatory T cell population.⁷⁴ In the murine model of doxorubicin (Adriamycin) nephropathy, the adoptive transfer of *Foxp3*-transduced T cells protected against kidney injury. Urinary protein excretion and serum creatinine were reduced, and there were significant decreases in glomerulosclerosis, tubular damage, and interstitial infiltrates.⁷⁵ These findings highlighted a complex interplay between effector/inflammatory cells and regulatory T cells in the setting of proteinuric chronic kidney diseases.

The modulatory role of Treg cells has also been recently supported by a study in a rat model of kidney ischemia-reperfusion injury (characterized by early proteinuria and increase in serum creatinine levels), after activation of the angiotensin II type 2 receptor (AT2R).⁷⁶ Treatment with the AT2R receptor agonist Compound 21 (C21) 2 hours before ischemic surgery attenuated postischemic kidney dysfunction and CD3⁺ and CD4⁺ T cell infiltration during the acute injury phase (2 hours postreperfusion). During the chronic phase (3–5 days), C21 treatment modulated the kidney microenvironment, as highlighted by increased CD4⁺ *Foxp3* cells and IL-10-secreting T CD4⁺ cells along with reduced levels of kidney injury molecule-1 and neutrophil gelatinase-associated lipocalin.⁷⁶ These findings pave the way to further explore AT2R-dependent kidney protective effects.

Proteasomal Processing of Albumin by Renal Dendritic Cells Generates Antigenic Peptides

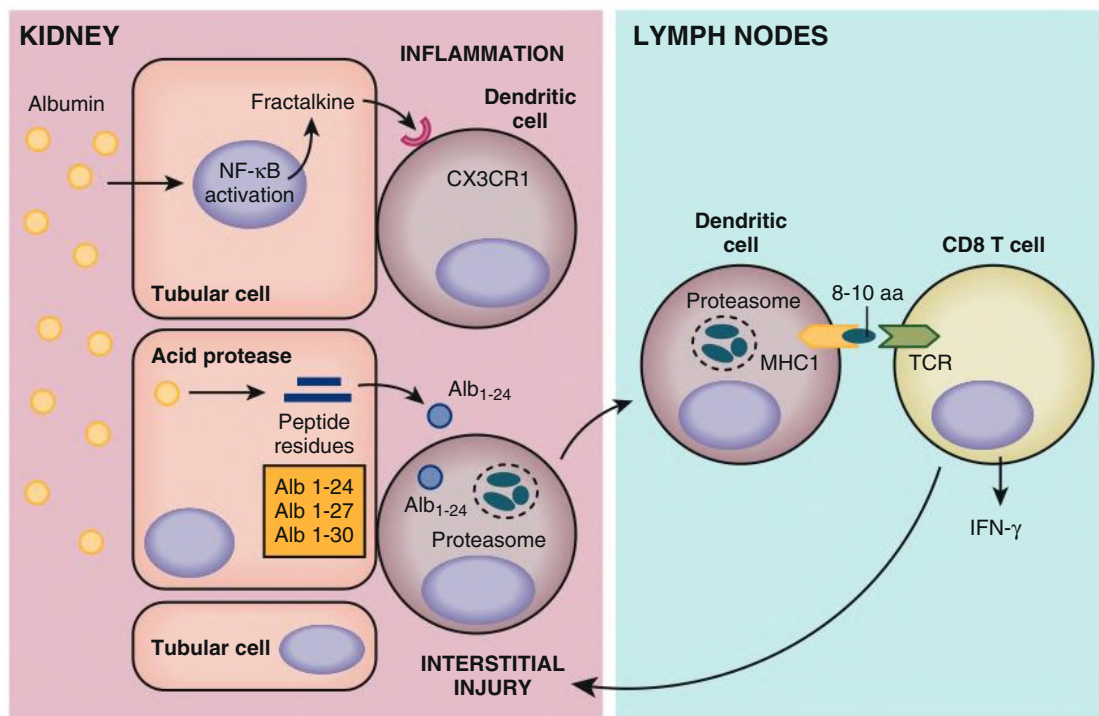


Fig. 81.6 Proteasomal Processing of Albumin by Kidney Dendritic Cells Generates Antigenic Peptides. Exposure of proximal tubular cells to excess autologous albumin, as in the case of proteinuric nephropathies, results in the formation of the N-terminal 24-residue fragment of albumin. This peptide is acquired by dendritic cells, where it is further processed by proteasome into antigenic peptides. Then dendritic cells move to regional lymph nodes, where they activate effector CD8⁺ T cells. These cells are then recruited in the kidney interstitium. *Alb*₁₋₂₄, N-terminal 24-residue fragment of albumin; *aa*, amino acids; *IFN-γ*, interferon-γ; *MHC*, major histocompatibility complex; *NF-κB*, nuclear factor-κB; *TCR*, T-cell receptor.

Fibroblast Activation and Extracellular Matrix Deposition

The process of interstitial fibrosis involves the accumulation of myofibroblasts and ECM proteins (Fig. 81.7).⁷⁷ Resident interstitial fibroblasts and myofibroblasts proliferate in response to macrophage-derived profibrogenic cytokines, and their number correlates with the subsequent formation of a scar.⁷⁷ These cells may derive from transdifferentiated tubular epithelial cells or pericytes of peritubular capillaries, a process promoted by profibrogenic cytokines, including TGF-β, expressed by macrophages.⁷⁸ Activated kidney fibroblasts may secrete chemokines that in turn may further attract macrophages and perpetuate interstitial injury.⁷⁹ Moreover, miRNAs are emerging as both downstream effectors of TGF-β–dependent regulation of the fibrogenic process.⁸⁰ TGF-β upregulates the expression of the profibrotic miR-21 in cultured proximal tubular epithelial cells via Smad3 signaling, both at the transcriptional and posttranscriptional level.⁸¹ A functional link between miR-192 and TGF-β–driven kidney fibrosis also has been documented, although the effect of TGF-β on miR-192 expression is not consistent across various studies.⁸⁰

Eventually, activated fibroblasts produce interstitial matrix components that contribute to interstitial collagen deposition and fibrosis. Increased tubulointerstitial fibrosis is a common feature of kidney injury and results from accumulation of ECM structural proteins. However, it is maintained by continuous remodeling through the proteolytic action of matrix metalloproteinases (MMPs) and the synthesis

of new proteins. MMPs are inhibited by tissue inhibitors of TIMPs. Of note, TIMP-3 levels are upregulated in patients with diabetic nephropathy, which anticipates more interstitial fibrosis.⁸²

A recent study suggests that an endoplasmic reticulum resident protein, thioredoxin domain containing 5 (TXNDC5), which was found to be specifically upregulated in collagen-secreting kidney fibroblasts, could be a key mediator in experimental kidney fibrosis.⁸³ Global deletion of *txn5* gene markedly attenuated the extent of kidney fibrosis in three mouse models of chronic kidney disease, that is, unilateral ureteral obstruction, unilateral ischemia-reperfusion injury, and folic acid nephropathy. Mechanistically, TXNDC5 augmented kidney fibrosis by promoting kidney fibroblast activation/proliferation and ECM production through posttranslational stabilization and upregulation of type I TGF-β receptor.⁸³

Accumulating evidence indicates that stimulation of AT2R may exert antifibrotic effects.⁸⁴ Indeed, AT2R-deficient mice were found to develop a more severe course of nephropathy and kidney fibrosis in models of diabetic kidney disease and 5/6 nephrectomy.^{85,86} Consistently, in a mouse model of type 1 diabetic nephropathy, treatment with the AT2R agonist C21 significantly reduced the expression of several inflammatory and profibrotic mediators, and upregulated collagen degradation induced by MMP-2 and MMP-9.⁸⁷

Chronic Hypoxia

One of the most important contributors to the development of tubulointerstitial fibrosis is chronic ischemia.⁸⁸ The production of

Mechanisms of Interstitial Damage Induced by Proteins

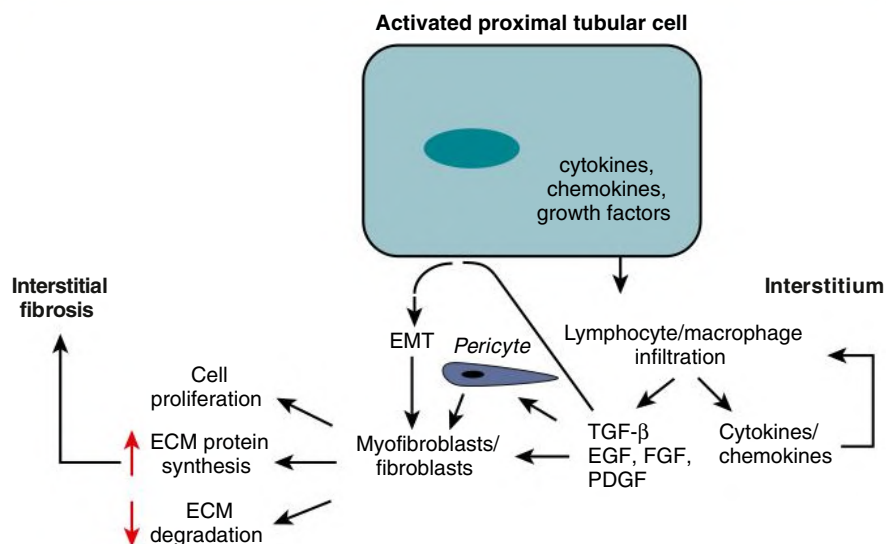


Fig. 81.7 Mechanisms of Interstitial Damage Induced by Proteins. Cytokines, chemokines, and growth factors are released from the activated tubule into the interstitium, where they contribute to recruit inflammatory cells and lymphocytes, causing progressive fibrosis. *ECM*, Extracellular matrix; *EGF*, epidermal growth factor; *EMT*, epithelial-to-mesenchymal transdifferentiation; *FGF*, fibroblast growth factor; *PDGF*, platelet-derived growth factor; *TGF-β*, transforming growth factor-β.

Ang II and the inhibition of the production of nitric oxide underlie chronic vasoconstriction, which may contribute to tissue ischemia and hypoxia.⁸⁹ In that regard, histologic studies of the human kidney have documented that there is often a loss of peritubular capillaries in areas of tubulointerstitial fibrosis.⁸⁹ Downregulation of VEGF may be functionally implicated in the progressive attrition of peritubular capillaries and tissue hypoxia. Moreover, given that the size of the interstitial compartment determines the diffusion distance between peritubular capillaries and tubular cells, interstitial fibrosis further impairs tubular oxygen supply. Focal reduction of capillary blood flow leading to the starvation of tubuli may underlie tubular atrophy and loss. Under these conditions the remaining tubules are subject to functional hypermetabolism, with increased oxygen consumption, which in turn creates an even more severely hypoxic environment in the kidney interstitium. Such hypoxia stimulates fibroblast proliferation and ECM production by tubular epithelial cells.⁹⁰

PRIMARY CHRONIC TUBULOINTERSTITIAL INJURY

Tubulointerstitial disease is common to all chronic progressive kidney diseases, irrespective of the initial trigger or site of injury. Once viewed as an inconsequential corollary to pathologic events that overwhelm glomeruli, tubulointerstitial disease is now recognized as a key and prominent factor in the progression of kidney disease. In addition to the setting of proteinuric nephropathies, inflammation and injury of the kidney interstitium also can occur in nonproteinuric kidney disease, as exemplified by primary chronic tubulointerstitial diseases. In this case the key event is the expression of local nephritogenic antigens. They are derived from kidney cells and tubular basement membranes (TBMs) or exogenous antigens processed by tubular cells. The case of molecules, including drugs, which may become nephritogenic antigens by acting as haptens or through molecular mimicry, is particularly unusual. A humoral response underlies rare cases of interstitial nephritis in which a portion of a drug (e.g., methicillin) may act as a

happen, bind to the TBM, and elicit anti-TBM antibodies.⁹¹ Among the numerous causes of primary chronic tubulointerstitial injury, drugs and toxins (see Table 65.1), in addition to genetic cystic diseases, are major players.

Analgesics and Nonsteroidal Antiinflammatory Drugs

Analgesic nephropathy is a specific form of kidney disease characterized by papillary necrosis and chronic interstitial nephritis, caused by prolonged and excessive consumption of analgesic mixtures. It is invariably caused by compound analgesic mixtures containing aspirin or antipyrine in combination with phenacetin, paracetamol, or salicylamide and caffeine or codeine, in popular over-the-counter proprietary mixtures.⁹² The mechanisms responsible for the kidney injury are incompletely understood. Phenacetin is metabolized to acetaminophen and to reactive intermediates that can injure cells, in part by lipid peroxidation.⁹³ These metabolites tend to accumulate in the medulla along the medullary osmotic gradient (created by the countercurrent system). As a result, the highest concentrations are seen at the papillary tip, the site of the initial vascular lesions.⁹⁴ The potentiating effect of aspirin, with both phenacetin and acetaminophen, may be related to two factors. (1) Acetaminophen undergoes oxidative metabolism by prostaglandin H synthase to reactive quinonimine, which is conjugated to glutathione. If acetaminophen is present alone, there is sufficient glutathione generated in the papillae to detoxify the reactive intermediate. However, if acetaminophen is ingested with aspirin, the aspirin is converted to salicylate, which becomes highly concentrated and depletes glutathione in both the cortex and papillae of the kidney. With the cellular glutathione depleted, the reactive metabolite of acetaminophen then produces lipid peroxides and arylation of tissue proteins, ultimately resulting in necrosis of the papillae.^{94,95} (2) Aspirin and other nonsteroidal antiinflammatory drugs suppress prostaglandin production by inhibiting cyclooxygenase enzymes. Kidney blood flow, particularly within the kidney medulla living at the edge of hypoxia, is highly dependent on systemic and local production of

vasodilatory prostaglandins. The final injury is therefore due to both the hemodynamic and the cytotoxic effects of these drugs, resulting in papillary necrosis and interstitial fibrosis.

Aristolochic Acid

In 1992 nephrologists in Belgium noted an increasing number of women presenting with kidney failure, often near end stage, after their exposure to a slimming regimen containing Chinese herbs.^{96,97} The main histologic lesion in human biopsy samples, which is located principally in the cortex, is extensive interstitial fibrosis with atrophy and loss of the tubules. Cellular infiltration of the interstitium is scarce. Important tryptase-positive mast cells were observed in the fibrotic areas in kidney biopsy samples.⁹⁸ Thickening of the walls of the interlobular and afferent arterioles resulted from endothelial cell swelling. The glomeruli are spared, relatively speaking, and immune deposits are not observed. These findings suggest that the primary lesions may be centered in the vessel walls, thereby leading to ischemia and interstitial fibrosis.⁹⁹

A plant nephrotoxin, aristolochic acid, has been proposed as the possible etiologic agent (see [Chapter 65](#)). Balkan endemic nephropathy, a chronic, familial, noninflammatory tubulointerstitial disease, most commonly seen in southeastern Europe, also has been potentially attributed to chronic exposure to dietary aristolochic acid (see [Chapter 65](#)).¹⁰⁰

Autosomal Dominant Polycystic Kidney Disease

Autosomal dominant polycystic kidney disease (ADPKD) is the most common monogenic kidney disorder, accounting for 8% to 10% of patients receiving renal replacement therapy for ESKD worldwide.¹⁰¹ In ADPKD patients, mutations in genes encoding for polycystin 1 or polycystin 2 lead to polycystin complex dysfunction. This results in reduced intracellular calcium concentration, leading to higher activity of adenylyl cyclase enzyme and upregulation of 3',5'-cyclic adenosine monophosphate (cAMP) levels.¹⁰² In turn, high intracellular cAMP levels lead to aberrant tubular epithelial cell proliferation and chloride-driven fluid secretion, the two key components of the process of cyst formation and growth in ADPKD.¹⁰³ In addition, activation of the Ser/Thr kinase mammalian target of rapamycin (mTOR) may

also contribute to cyst expansion.¹⁰⁴ Uncontrolled cyst growth results in crowding of adjacent nephrons, injury of normal kidney parenchyma, interstitial inflammation and fibrosis, and eventually, substantial enlargement of the kidneys and progressive kidney failure.^{105,106} It is assumed that kidney function starts to decline only after several decades, whereas cyst formation and growth already starts in utero. This preservation of GFR in early-stage ADPKD has been posited to be caused by compensatory hyperfiltration of remnant nephrons that have not yet been lost due to disease progression, a phenomenon referred to as glomerular hyperfiltration.¹⁰⁷ In the long term, however, this maladaptive response to reduced nephron number may translate into accelerated loss of residual hyperfunctioning nephrons and relentless progression to kidney failure. Indeed, evidence points to glomerular hyperfiltration as a risk factor for ADPKD progression. In particular, in 180 children with ADPKD, glomerular hyperfiltration (defined as creatinine clearance ≥ 140 mL/min/1.73 m²) was associated with an increased rate of kidney growth and faster GFR decline.¹⁰⁸ Similarly, among 91 adult ADPKD patients, glomerular hyperfiltration at baseline (140 mL/min/1.73 m²) was associated with higher GFR loss during the 15-year follow-up.¹⁰⁹ Therefore, the combination of cyst growth, tubulointerstitial injury, and maladaptive hyperfiltration of the remaining nephrons eventually results in ADPKD progression.

Insights into the pathophysiology of ADPKD have helped develop novel disease-modifying treatments (see [Chapter 46](#)).^{110,111}

CONCLUSIONS

There is now clear evidence that, rather than simply a marker of damage, abnormal ultrafiltration of proteins can be toxic to the kidney and incites complex pathways and mediators that target glomerular, tubular, and interstitial cells, eventually promoting kidney disease progression to ESKD. Moreover, tubulointerstitial injury is also common in nonproteinuric kidney diseases and is critical for their progression to ESKD. More insights in the pathophysiologic mechanisms of progressive proteinuric and nonproteinuric kidney diseases will be instrumental to search for novel biomarkers that allow intervention in the earlier stages of CKD and may open the way to new treatments.

SELF-ASSESSMENT QUESTIONS

- What is the pathologic podocyte response to protein trafficking throughout the glomerulus?
 - Podocytes start to proliferate.
 - Podocytes are damaged by excessive protein load and uptake of albumin.
 - Podocytes adhere to Bowman's capsule.
- What is the main consequence of podocyte loss on endothelial cells during chronic kidney injury?
 - Endothelial cell apoptosis
 - Endothelial cell proliferation
 - Increase of vascularization of the kidney
- What is the role of resident dendritic cells in the kidney during proteinuric diseases?
 - Dendritic cells proliferate.
 - Dendritic cells generate albumin peptides that activate CD8⁺ T cells.
 - Dendritic cells undergo apoptosis.
- How does the progressive enlargement of kidney cysts in autosomal dominant polycystic kidney disease develop?
 - It develops from mesangial cell proliferation.
 - It develops from endothelial cell proliferation.
 - It develops from tubular cell proliferation and fluid transport into the cyst cavity.

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